

MANAS AGRAWAL

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EDUCATION

Northeastern University

Masters of Science – Computer Science | GPA: 3.86

Boston, MA

Sep 2024 – Dec 2026

- Graduate Teaching Assistant for Fundamentals of Software Engineering (CS 4530)
- Researched Multi-Agent LLM architectures, for e2e SDLC automation, across accuracy, cost efficiency

WORK EXPERIENCE

Hydrow

Software Engineer Intern

Boston, MA

Jan 2026 – Aug 2026

- Kept every user-facing endpoint under 25 ms at production scale for several product features end-to-end — schema, APIs, tests, and rollout — including the strength-fitness app's progress graphs.
- Cut a real-time badge eligibility check from ~2,000 ms to near-zero and an event-loop-blocking graph computation to 0.7 ms by pushing data-heavy work into SQL with materialized views and Redis caching.
- Restored data correctness across ~123K historical records via a bounded, dry-run-verified backfill migration, and kept large-table reads scale-safe with cursor-based pagination.
- Built an AI-assisted development workflow on Claude Code that automates the coding, testing, and self-review stages of the SDLC while keeping me in the loop for planning and final review, backed by a persistent knowledge base of accumulated review standards.
- Eliminated dead-end workout recommendations across all equipment and difficulty combinations by building a multi-pass SQL pipeline that relaxes constraints stepwise until it returns a viable workout.
- Reduced new-badge rollout from a coordinated frontend + backend change to backend-only across 21 badge types by shipping a schema-driven API that generates admin forms from validation schemas, plus 5+ real-time badge checkers.
- Enabled per-segment content delivery without redeployes by designing a pluggable A/B framework with a master/leaf tile mechanism targeted on churn score and behavioral segment.

canAssist × Northeastern University

Product Engineer & Scrum Master, Research Capstone

Boston, MA

May 2026 – Aug 2026

- Delivered a working assistive-technology product to a real client by leading a 5-person team through requirements gathering, sprint ceremonies, and demos across a 4-month capstone.
- Built the NestJS/PostgreSQL/Redis backend instrumented with my own open-source boilerplate and telemetry package.

Studio Graphene

Forward Deployed Engineer

Gurgaon, India

Nov 2020 – Jul 2024

- Boosted engineering velocity 23% by building a serverless analytics platform on AWS Lambda that pinpointed PR wait times, build failures, and blocked dependencies.
- Served 2M+ daily requests against a 500GB+ catalog by designing a Node.js/TypeScript REST backend with PostgreSQL partitioning/indexing and real-time cross-region inventory sync.
- Reduced response latency ~40% by moving ingestion and processing to event-driven microservices on AWS SQS.

OPEN SOURCE PROJECTS

[Nodejs Observability](#) | *Nodejs, TypeScript, AWS X-Ray, npm, OpenTelemetry, cloudwatch, Jaeger, Prometheus*

- A vendor-agnostic OpenTelemetry package with pluggable presets for AWS, Honeycomb, Jaeger, Tempo, or any OTLP-native backend. Ships traces, metrics, and logs through a single sidecar collector with cost-mindful defaults. Reduces mean debugging time by ~60%
- It has over 1500 downloads on npm

TECHNICAL SKILLS

Languages:

JavaScript, TypeScript, Java, C++, Python

Frameworks:

Node.js, Express.js, NestJS, Django, Laravel, React

Database:

SQL, Postgres, MySQL, NoSQL, MongoDB, Elasticsearch, Amazon DynamoDB, Firebase

Cloud & DevOps:

Cloud Computing, AWS (Lambda, S3, SQS, DLQ, X-Ray), Docker, CI/CD, cloud infrastructure

Others:

Git, Sentry, OpenTelemetry, linux, REST APIs, web architecture, frontend, agile, SDLC